Forced Alignment Using Montreal Forced Aligner (MFA)

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Project: Speech-to-Speech Machine Translation Internship – Assignment 1

1. Objective

The aim of this assignment was to perform forced alignment between audio recordings and their corresponding transcripts using **Montreal Forced Aligner (MFA)**. Forced alignment identifies the start and end times of words and phonemes in speech given a transcript.

2. Methodology

1. Environment Setup
    a. Created a Conda environment named aligner.
    b. Installed Montreal Forced Aligner via Conda.

2. Data Preparation
    a. Used the provided dataset of .wav audio files and .txt transcripts.
    b. Dataset organized according to MFA's required format.

3. Model and Dictionary
    a. Downloaded pretrained acoustic model: english_us_arpa.zip
    b. Downloaded pretrained dictionary: english_us_arpa.dict

4. Forced Alignment Execution
    a. Ran the alignment using the command:
    mfa align --clean --overwrite "dataset" "english_us_arpa.dict" "english_us_arpa.zip" "outputs"
    b. Output: TextGrid files marking word and phoneme boundaries.

5. Output Inspection
    a. Opened TextGrid files in **Praat** software.
    b. Observed phoneme and word boundaries.
    c. Took screenshots from Praat to visualize alignments.

3. Sample Alignment

File: ISLE_SESS0131_BLOCKD02_01_sprt1.TextGrid

Words:
0.0 – 0.44 (silence)
0.44 – 0.53 i
0.53 – 0.92 said
0.92 – 1.33 white
1.33 – 1.48 (silence)
1.48 – 1.8 not
1.8 – 2.25 bait
2.25 – 4.125 (silence)

Phonemes:
0.0 – 0.44 (silence)
0.44 – 0.53 AY1
0.53 – 0.71 S
0.71 – 0.79 EH1
0.79 – 0.92 D
0.92 – 1.03 W
1.03 – 1.17 AY1
1.17 – 1.33 T
1.33 – 1.48 (silence)
1.48 – 1.55 N
1.55 – 1.62 AA1
1.62 – 1.8 T
1.8 – 1.84 B
1.84 – 2.05 EY1
2.05 – 2.25 T
2.25 – 4.125 (silence)

4. Automation (Extra Credit Work)

To enhance the workflow and earn extra credit, the entire forced alignment process was automated using a Python script and a batch file.
This approach eliminates the need to manually type long commands and ensures consistent results across multiple systems.

Automation Script
A Python script named automate_alignment.py was created inside the scripts/ folder.
It dynamically locates the dataset, models, and output directories relative to its location, and runs the full alignment command automatically.

Key Features:

- Automatically detects paths (non-static).
- Executes MFA with --clean and --overwrite flags.
- Ensures reproducibility and efficiency.
- Compatible with all systems using dynamic path detection.

Batch File
A batch file named run_alignment.bat was created to simplify execution.
It activates the conda environment and runs the Python script automatically.
Users can simply double-click the batch file to start the alignment.

Usage
To run the automated alignment:

- Ensure the Conda environment (aligner) is set up with MFA.
- Double-click run_alignment.bat.
- The aligned TextGrid files will be saved in the outputs/ folder.

Automation Workflow Summary:
run_alignment.bat → activates aligner environment → runs automate_alignment.py → executes MFA alignment → saves aligned TextGrids in outputs/

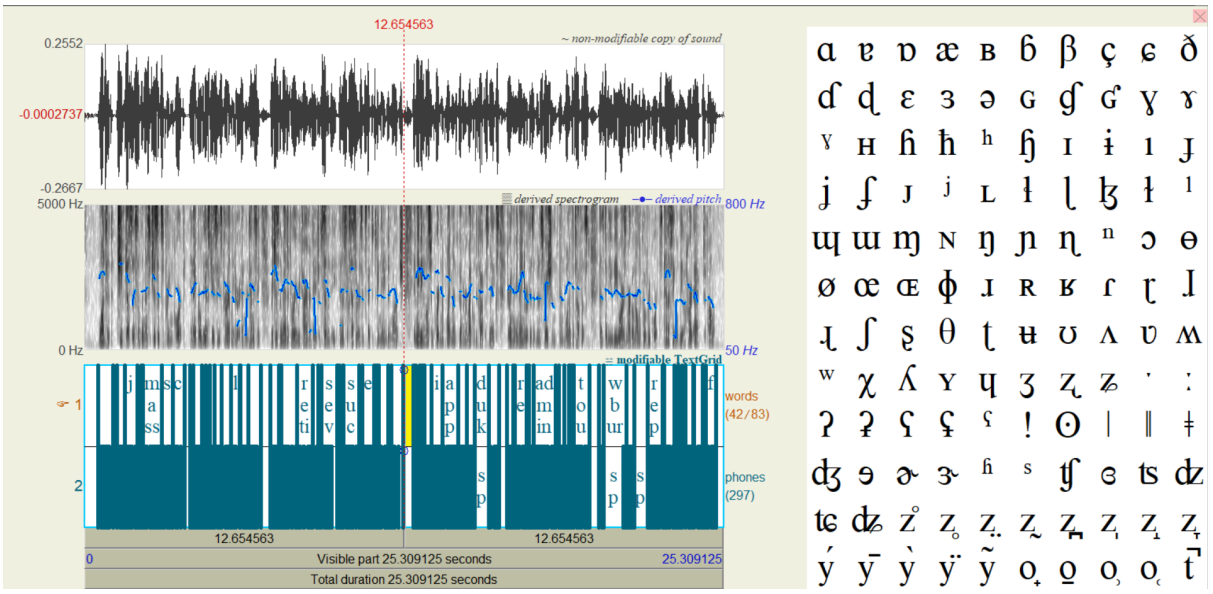
5. Observations

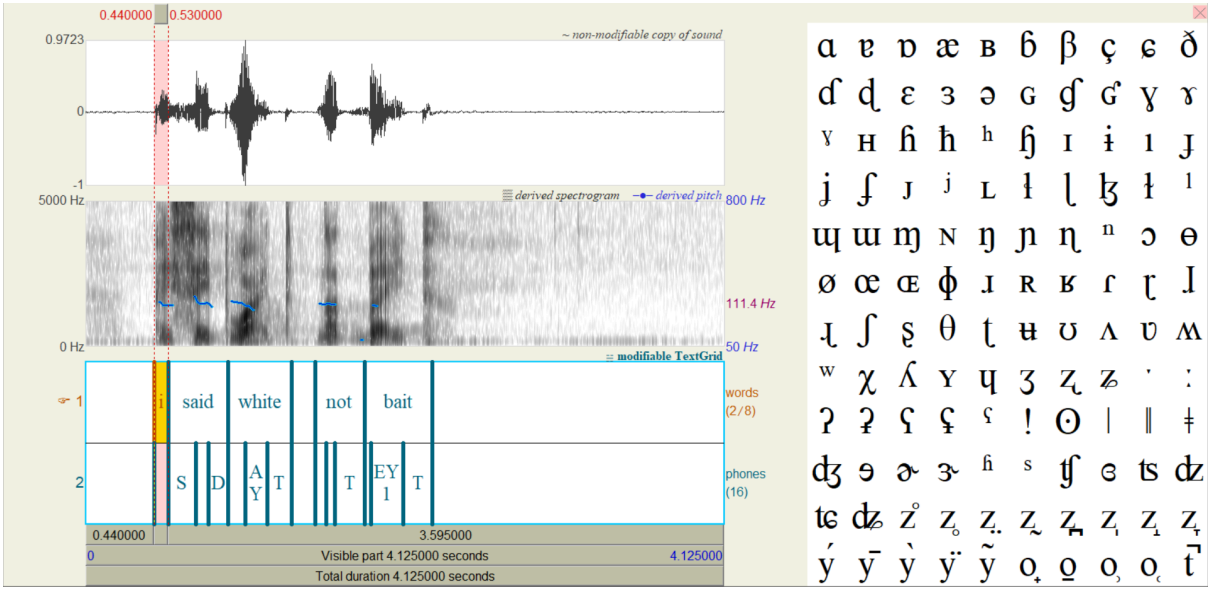
1. MFA successfully aligned all audio files with high accuracy.
2. Minor warnings appeared due to processing a single speaker, but alignment quality remained unaffected.
3. The automation script provided a clean and efficient way to repeat the alignment process.
4. Praat allowed clear visualization of word and phoneme boundaries.

5. Screenshots from Praat

Screenshots of TextGrid visualizations were captured using Praat.
These screenshots illustrate how the word and phoneme intervals are aligned with the audio waveform.

Screenshots are also included in the GitHub repository under the folder: praat_screenshots.





6. Conclusion

The assignment demonstrates how MFA can be effectively used for forced alignment in speech-to-text tasks. Using pretrained models and dictionaries simplifies alignment and allows easy visualization of phonetic and word-level timing information.

GitHub Repository:- https://github.com/skfareeduddin/mfa_forced_alignment